GOVERNED MODEL REFERENCE

AppSheet Data Model Overview

Money database | Loan Manager ผู้จัดการสินเชื่อ | Loan Business Insights

Descriptive companion to AppSheet_Data_Dictionary.xlsx

Control	Value
Document role	Architecture, governance, purposes, assumptions, and review guidance
Technical inventory	AppSheet_Data_Dictionary.xlsx
Source baseline	AppSheet_Data_Model_Reference.docx - configuration dated 2026-08-29
Conversion date	30 August 2026 (Asia/Singapore)
Reconciliation status	Pending user reconciliation with live AppSheet configuration
Maintenance model	Excel is authoritative for row-level technical inventory; Word explains context and governance

Conversion boundary This split reproduces the existing Word baseline. It does not assert that the baseline still matches the live AppSheet database or either app as of the conversion date.

1. Purpose, authority, and maintenance

This overview explains how the shared database, the OLTP Operations app, and the OLAP Reporting app fit together. Detailed tables, columns, expressions, application variances, slices, and slice-column membership are maintained in the companion Excel workbook.

1.1 Authority and status

System authority: The AppSheet database and live app editors remain authoritative.

Conversion authority: The content in this pair was converted from the existing Word baseline only.

Reconciliation status: Pending user reconciliation; no AppSheet setup was opened or compared during this conversion.

Sensitive data: The reference describes configuration only and contains no production records, borrower documents, bank details, or credentials.

1.2 Division of responsibility

Artifact	Role	Contents	Update trigger
Word overview	Narrative	Architecture, ownership, purposes, assumptions, risks, and maintenance workflow	Update when meaning, ownership, architecture, or governance changes
Excel dictionary	Technical inventory	Tables, columns, expressions, relationships, slices, slice columns, and app variances	Update for every data-model or slice change
Excel change log	Audit trail	What changed, when, why, and whether it was reconciled with AppSheet	Append one row for every accepted model revision

1.3 Maintenance procedure

1. **Inspect:** Review the affected source table and both app definitions before changing the reference.
2. **Update Excel:** Edit the affected inventory rows first and preserve stable object names and column order.
3. **Update Word:** Revise this overview only when architecture, purpose, ownership, assumptions, risk, or validation guidance changes.
4. **Record:** Append a dated Change Log row in the workbook, including reconciliation status and source baseline.
5. **Verify:** Check filters, formulas as text, relationships, slice membership, and cross-document consistency before accepting the revision.

Maintenance rule Do not duplicate the full technical dictionary in Word. Excel is the row-level source of truth for this reference pair; Word supplies context.

2. Architecture overview

Component	Named asset	Current scope	Ownership / purpose
Money database	Operational system of record	10 AppSheet Database tables	Physical keys, stored fields, and referential integrity
OLTP Operations	Loan Manager ผู้จัดการสินเชื่อ	6 loaded tables; 2 slices	Data entry, validation, collection workflow, and lightweight operational indicators
OLAP Reporting	Loan Business Insights	11 loaded tables; 11 slices	Historical analytics, partner analysis, loan assessment, and curated reporting feeds
External helper	Calendar Help Table / Sheet1	1 Google Sheets table	Date-spine calculations for the OLAP loan-assessment model

The OLTP and OLAP apps share the same Money database. Application-specific variance is primarily expressed through loaded tables, virtual columns, and curated slices. The Calendar helper remains an external Google Sheets-backed input to the OLAP model.

3. Table catalog

The companion workbook contains the complete column-level dictionary. This table remains in Word because source ownership and table purpose are architectural facts.

Table	Source	Loaded in	Physical	OLTP VCs	OLAP VCs	Purpose
Borrowers	Money database	OLTP + OLAP	6	24	50	Borrower master records and borrower-level operational or portfolio indicators.
Calendar	Calendar Help Table / Sheet1	OLAP only	2	0	22	Google Sheets-backed date spine used by OLAP loan-assessment projections; not part of Money database.
Cash Pool Contributions	Money database	OLAP only	7	0	2	Partner capital-contribution movements that build the lendable cash pool.
Charges	Money database	OLTP + OLAP	7	15	18	Scheduled amounts due and collection state at charge level, including principal and interest allocation.
Daily Analytics	Money database	OLAP only	17	0	6	Persisted daily portfolio snapshot fact table for historical reporting and governed feeds.
Loan Assessment	Money database	OLAP only	6	0	16	Input/control record for evaluating a proposed loan against projected coverage and return.
Loans	Money database	OLTP + OLAP	19	19	25	Loan contracts, commercial terms, lifecycle state, balances, and loan-level risk metrics.
Partners	Money database	OLTP + OLAP	5	0	12	Partner master records used for capital participation and settlement reporting.
Repayments	Money database	OLTP + OLAP	9	4	13	Posted payment transactions and their allocation to loans and charges.
Settlements	Money database	OLAP only	8	0	0	Auditable partner settlement records and settlement status.
Statistics	Money database	OLTP + OLAP	3	23	36	Small KPI/control table used to expose app-level operational and reporting aggregates.

Physical count includes the AppSheet or Sheets row ordinal where exposed. Row ID is the documented stable key for AppSheet Database tables; Calendar uses Date as its key.

4. Relationship model

The detailed workbook records the same direct Ref links in a filterable table. Reverse-reference virtual columns remain part of the Columns sheet.

Child table	Ref column	Parent table	Cardinality	Is a part of?
Calendar	Assessment Borrower	Borrowers	Borrowers 1 -> many Calendar	Independent child lifecycle
Cash Pool Contributions	Ref Partner	Partners	Partners 1 -> many Cash Pool Contributions	Independent child lifecycle
Charges	Ref Loans	Loans	Loans 1 -> many Charges	Child rows are marked part-of parent
Loan Assessment	Ref Borrower	Borrowers	Borrowers 1 -> many Loan Assessment	Independent child lifecycle
Loans	Ref Borrowers	Borrowers	Borrowers 1 -> many Loans	Child rows are marked part-of parent
Repayments	Ref Loans	Loans	Loans 1 -> many Repayments	Independent child lifecycle
Repayments	Ref Charges	Charges	Charges 1 -> many Repayments	Child rows are marked part-of parent
Settlements	Ref Partner	Partners	Partners 1 -> many Settlements	Independent child lifecycle

Cascade caution Rows marked Is a part of? may follow the referenced parent lifecycle. Financially material corrections should remain auditable and should not rely on routine physical deletion.

5. Slice catalog

App	Slice	Source	Row filter	Mode	Cols	Purpose
OLTP	Todays Collection Board	Borrowers	ISNOTBLANK([Daily Collection Status])	Read-only	10	Read-only borrower workboard restricted to borrowers with a current collection status.
OLTP	Payment Collection	Charges	OR([Payment Status] = "รอชำระ", [Payment Status] = "ชำระบางส่วน", AND([Payment Status] = "ชำระแล้ว", [Payment Date] = TODAY()))	Updates, Adds, Deletes	22	Operational charge queue containing pending or partially paid items plus items completed today.
OLAP	Borrowers with Outstanding Loans	Borrowers	[Total Outstanding Principal] > 0	Read-only	56	Read-only OLAP subset of borrowers whose outstanding principal is greater than zero.
OLAP	Profit Contribution 30D	Borrowers	[Profit Last 30 Days] > 0	Read-only	56	Read-only OLAP subset of borrowers contributing positive interest profit in the last 30 days.
OLAP	Reporting Borrowers	Borrowers	TRUE	Read-only	40	Curated, read-only borrower feed for downstream reporting and dashboard reconciliation.
OLAP	Reporting Charges	Charges	TRUE	Read-only	20	Curated, read-only charge feed with due, paid, remaining, lateness, and borrower fields.

App	Slice	Source	Row filter	Mode	Cols	Purpose
OLAP	Daily Analytics 7 Days	Daily Analytics	[Snapshot Date] >= NOW() - 7	Read-only	23	Rolling seven-day daily-analytics window for recent trend analysis.
OLAP	Daily Analytics 30 Days	Daily Analytics	[Snapshot Date] >= NOW() - 30	Read-only	23	Rolling 30-day daily-analytics window for monthly and near-term trend analysis.
OLAP	Daily Analytics 90 Days	Daily Analytics	[Snapshot Date] >= NOW() - 90	Read-only	23	Rolling 90-day daily-analytics window for medium-term portfolio trends.
OLAP	Reporting Daily Analytics	Daily Analytics	TRUE	Read-only	13	Curated, read-only daily snapshot feed for governed downstream reporting.
OLAP	Reporting Loans	Loans	TRUE	Read-only	30	Curated, read-only loan feed containing contract, balance, rate, age, and default fields.
OLAP	Reporting Repayments	Repayments	TRUE	Read-only	8	Curated, read-only posted-payment feed with loan and charge references.
OLAP	Reporting Statistics	Statistics	TRUE	Read-only	17	Curated, read-only KPI feed for governed dashboard totals.

Exact slice-column membership and order are maintained only in the workbook's Slice Columns sheet.

6. Financial-model assumptions retained from the baseline

The following notes are carried forward verbatim in substance from the existing Word baseline. They have not been re-tested during this conversion.

The existing KPI names are retained. Interpret the monthly forecast using the following assumptions:

- Projected Monthly Profit equals Realized Profit MTD plus Today's Outstanding Interest plus Forecast Remaining Interest.
- Realized Profit MTD is based on Repayments[Interest Paid] whose Payment Date falls in the current calendar month. Under the current classification, first-day transfer fees recorded through Charge interest and negative default adjustments recorded through Interest Paid flow into this amount.
- Today's Outstanding Interest is not limited to today or the current month. It sums Charges[Interest Remaining] across all open Charges and may therefore include arrears originating before the current month.
- For daily-interest loans, Forecast Remaining Interest includes accrued but uncharged interest since the latest Charge plus interest for days after today through month-end. When a Charge is generated or paid, the amount should move between uncharged accrual, Charge interest remaining, and realized interest without intentional double counting.
- The projection assumes all outstanding interest is collectible; it does not apply delinquency, default-probability, or collection-probability weighting.
- Daily-interest accuracy assumes Current Daily Interest is maintained in accordance with the loan contract, including after any principal change; the payment workflow does not automatically recalculate that stored value.
- For fixed-term loans, interest due later in the current month remains forecast until the due date, when the 00:05 bot is expected to create the Charge. A delayed or failed due-date bot can temporarily omit that fixed interest from both forecast and outstanding Charge interest.

Validation status Treat these assumptions as pending confirmation until the user reconciles the converted reference with the current AppSheet formulas, bots, and business rules.

7. Known implications and follow-up checks

Area	Implication
OLTP/OLAP variance	Shared physical tables have materially different VC layers. Changes to a shared field must be reviewed in both app dictionaries even when only one app appears to use it.
Reporting ownership	OLAP-only VCs and Reporting * slices are the preferred governed feed surface. Do not move their heavy aggregation into OLTP without a performance assessment.
Calendar source	Calendar is not in Money database. Its key/date population and refresh behavior depend on Calendar Help Table / Sheet1 and must be checked separately.
Slice comments	All inspected slice descriptive-comment fields were blank. This document supplies purposes, but future changes should also populate AppSheet descriptive comments where practical.
Delete permissions	OLTP Payment Collection currently permits updates, adds, and deletes. Any production correction to financial rows should remain auditable and avoid routine physical deletion.
Dashboard reconciliation	Reporting slice totals should be reconciled to their governed base grains and refresh cutoff before Looker Studio changes are accepted.

7.1 Manual validation checklist

Schema: Confirm the Money database still reports 10 tables and that each physical field matches the detailed dictionary.

Relationships: Open every listed Ref Type Details panel and confirm source table plus Is a part of? state.

App variance: Confirm the OLTP and OLAP column counts shown at each table heading match this document.

Slices: Confirm row filters, included-column order, and update mode for all 13 slices.

Permissions: Verify reporting slices remain read-only and operational update access matches intended roles.

Financial logic: Sample posted, partial, fully paid, overdue, reversed/adjusted, and defaulted scenarios against the documented VC dependencies.

Reconciliation: Reconcile OLAP Reporting slice totals to Money database at a defined as-of time before dashboard use.

8. Companion workbook map

Worksheet	Purpose
README	Status, maintenance rules, scope, and formula-driven inventory counts
Tables	One row per source table with app loading, physical/VC counts, and purpose
Columns	One row per unioned column definition, including type, availability, flags, purpose, and expression
Relationships	One row per direct Ref relationship and part-of behavior
Slices	One row per OLTP/OLAP slice with filter, update mode, purpose, and column count
Slice Columns	One row per included slice column, preserving display order
App Variances	Tables and columns present in only one app
Change Log	Append-only maintenance history and reconciliation status

Next action Reconcile the workbook against the live AppSheet database and app editors, then update the Change Log status from Pending to Reconciled.